

Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT Sparse PMUs can recover a requested voltage field while leaving the physical interventions that produced it ambiguous. We formulate these tasks with a hybrid differential-algebraic model that separates physical interventions from measurement integrity, and we distinguish functional state observability from finite-horizon event-source diagnosability. The evaluated load-intervention likelihood assigns posterior mass to event cardinality, support, and signed amplitude over a closed bank of 137 hypotheses: no event, 16 single sources, and all 120 two-source supports. Its quadratic response manifold includes source self-curvature and pairwise interaction. On 3488 fresh nonlinear trajectories and 10564 frozen noise realizations, exact support Top-1 is 24.4% for weak-weak pairs, 66.2% for weak-strong pairs, 98.8% for moderate pairs, and 100.0% for finite pairs. Cardinality accuracy in the weak-weak regime is 37.4%, showing that detecting two active sources can precede identifying them. A separate causal estimator reconstructs 31 hidden bus voltages from eight voltage/current PMUs with 0.009634% mean error, while physical recentering removes 97.4%–98.3% of an operating-point error gap. Retrospective geometry links profiled distance to weak-case support recovery, but the 120-frame diagnostic remains limited to one operating point. The evidence supports matched-simulator state reconstruction and load-intervention inference, not a validated multi-family joint estimator, changing PMU sets, unknown onset, or field transfer.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. A disturbance may begin outside the measured region, alter hidden voltages, and propagate to the available devices. Missing frames and corrupted channels create a second ambiguity because they can resemble physical transients. Diagnosis must therefore separate three questions: what is the network state, which event occurred, and where did each applicable physical or measurement event originate?

Dynamic estimators combine network models with synchrophasors [2], [3]. Sparse-PMU classifiers identify events and faulted assets [4], [5]. Other learning methods reconstruct hidden trajectories before localization or estimate state

and attack location from partial observations [6], [7]. These methods solve different inverse problems. The unresolved case considered here has an under-observed state, concurrent physical and measurement-integrity changes, source assets absent from labeled training, and sensing geometries that may render two sources indistinguishable.

The distinction is operational, not semantic. A generation change can be visible throughout the network while its source remains ambiguous. A dropped PMU identifies an integrity location directly but removes information needed to localize a simultaneous physical event. Likewise, a state estimate can have small point error while the intervention that produced it is not uniquely identifiable. A single multiclass label conceals these differences and produces accuracy values whose denominators do not match the claimed task.

A physical candidate contains an event family, a sparse

source support, onset, and signed severity. Measurement availability and corruption remain separate variables. Each candidate generates a counterfactual DAE response, while candidate comparison profiles admissible state and integrity nuisance directions. Specializing descriptor-model distinguishability and matched-subspace ideas [8], [9], the remaining weighted distance measures local source separation. A zero distance identifies an intersecting pair of response manifolds; a positive but small distance identifies a decision that is structurally distinct yet noise and horizon limited.

Three frozen contracts supply complementary evidence. The PowerDynamics state study reconstructs 31 hidden complex bus voltages from eight voltage/current PMUs and tests operating-point mismatch. The principal event study evaluates 137 load-intervention hypotheses— H_0 , 16 single buses, and 120 simultaneous pairs—using a quadratic physical response manifold and normalized Bayesian evidence, without a fitted source-label head. A retrospective weak-regime audit and a longer-horizon diagnostic probe why support decisions fail. The separate ANDES study preserves the earlier physics-guided hierarchical baseline on eight abnormal labels plus normal operation. The simulators, targets, and splits remain separate; their metrics are never pooled.

The contributions are:

- 1) a hybrid DAE problem formulation that maps eight historical labels to composable physical-intervention and measurement-integrity variables, while representing simultaneous load sources by a sparse support rather than a new class label;
- 2) a finite-horizon analysis that separates target-state estimability from sparse event-support injectivity, nuisance-profiled source separation, and the conditioning gained from additional measurements or observation time; and
- 3) a fresh 137-hypothesis Bayesian confirmation over all zero-, one-, and two-source load supports, together with frozen state-reconstruction, operating-point recentering, weak-regime geometry, and source-holdout evidence.

The empirical boundary is sharp. The state estimator reaches 0.009634% mean hidden-voltage HVRE, while physical recentering closes 97.4%–98.3% of the operating-point error gap. In the prospective simultaneous-load test, exact support Top-1 rises from 24.4% for weak–weak events to 100.0% for finite events; weak cardinality is often resolved before exact support. The retrospective profiled distance predicts weak-case exact recovery far better than a support-only score. A 120-frame diagnostic finds no equilibrium-degenerate source pair in the frozen local model, but near-collinearity and incomplete independent validation prevent a general no-ambiguity claim. Physical candidate inference is therefore supported for matched load interventions with at most two sources, not for the full eight-family joint problem.

II. PRIOR WORK AND RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with synchronized measurements; model uncertainty and measurement availability remain persistent constraints [2]. Robust filters address non-Gaussian measurement errors [10], [11], whereas joint input/state estimators recover exogenous inputs in linear systems [12], nonlinear DAEs [3], [13], and event-triggered or distributed power networks [14], [15]. These estimators do not define a multi-family physical-source bank with a simultaneous integrity state.

Learning methods reconstruct states from sparse measurements by encoding graph structure or power-flow relations. Examples combine graph learning with physics-based estimates [16], outlier-resistant estimation and outage identification [17], or topology-aware reconstruction under PMU failures and bad data [18]. Robust reconstruction does not, by itself, identify the physical source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification uses subspaces, engineered features, transfer learning, and graph models [19]–[22]. Label-light field-stream detection and scarce-label transfer target event presence or family [23], [24]; physical-source recovery is distinct. Prior localization spans joint classification/localization [4], sparse-PMU fault, generation-loss, and outage methods [5], [25]–[27], group-sparse physical dictionaries [28], and topology-constrained graph losses [29].

Model-based outage inference provides the closest classical comparison. Phasor angles support line-outage detection [30]; uncertain pre-event state can be marginalized jointly with outage identity and sensor placement [31]; and sequential likelihoods handle stochastic operating variation [32], [33]. Matched-subspace detectors profile linear nuisance through a generalized likelihood ratio [9]. These are the likelihood and nuisance-elimination precedents used here. Our load contract instead uses a sparse voltage/current response dictionary and known onset; it is not a quickest-detection comparison.

Open-set recognition differs from source holdout. Siamese boundaries and generative evidence reject events outside learned classes [34]–[36]. Source generalization instead ranks a physically enumerable but unlabeled asset as a counterfactual source.

The pmuBAGE collection supports standardized analytics but does not certify source holdout [37]. Integrity is a separate variable: robust recovery distinguishes excursions from outliers [38], while utility examples include missing, locked, quantized, and timing-affected synchrophasors [39].

The unresolved issue is the evaluation contract. New replicas from trained assets measure interpolation, not source generalization; nuisance variation can also alter conclusions [40]. Reconstruction followed by localization is prior art [6], [7], but not under the combined held-source, diagnosability, integrity, and mismatch tests used here.

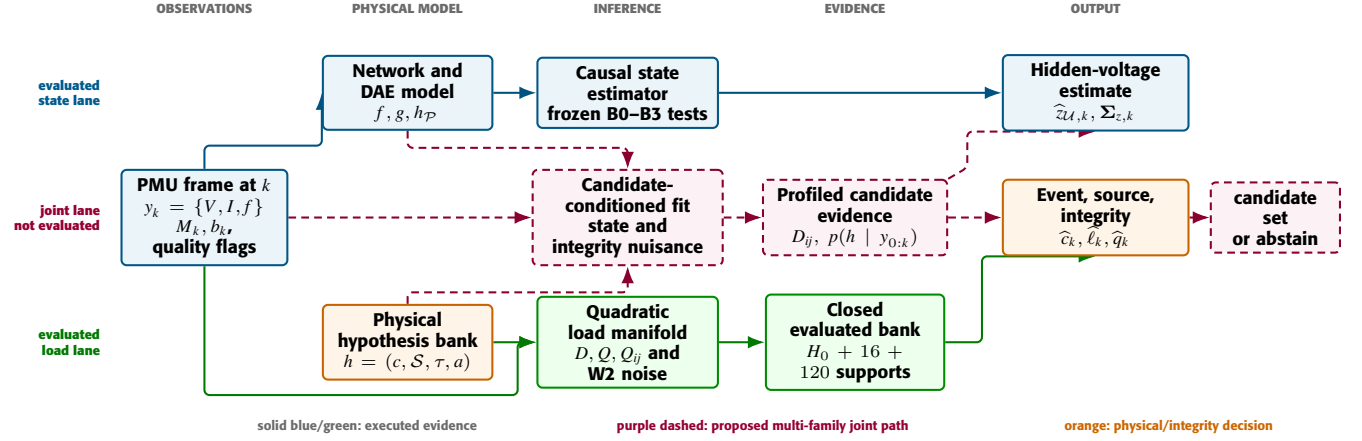


FIGURE 1. Inference path and validation boundary. The three lanes separate the evaluated causal state estimator, the proposed multi-family candidate-conditioned route, and the evaluated zero-, one-, and two-source load-intervention case. Solid blue and green boxes denote executed evidence; purple dashed boxes denote the multi-family nuisance fit and calibrated abstention path that still require a common experiment.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [41], [42]; nonlinear extensions require the observation fibers to determine the requested functional locally, not merely a favorable Jacobian at one point [43]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [44]; fault detectability asks whether a fault can be separated while disturbances are decoupled [45]. Sparse strong observability extends the question to corrupted sensor sets [46], [47], and attack identification characterizes when adversarial input supports are distinguishable from system behavior [48]. Eriksson *et al.* quantify pairwise diagnosability for stochastic linear descriptor models [8]. Likelihood inference with singular first-order information also has established nonregular asymptotics [49]; therefore, a vanishing projected tangent is not by itself a universal impossibility result. PMU placement based on empirical observability Gramians treats sensor value as task dependent [50], [51]. For event sources, Jamei *et al.* derive placement-dependent fault-localization limits [52]; Conte *et al.* partition a radial grid into indistinguishable line clusters [53]; and Yildiz and Abur optimize sparse placement for event detection and identification [54]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

D. REMAINING TECHNICAL GAP

Table 1 separates the capabilities needed for the combined problem. A source-agnostic physical hypothesis must be evaluated while the unobserved trajectory is estimated and measurement failures remain possible. The sensing geometry must expose indistinguishable sources before a classifier is trusted. Evaluation must then exclude the physical source asset from all source-labeled fitting. Under this protocol,

learning is useful only when it corrects model discrepancy or reduces candidate-search cost without memorizing asset labels.

The available evidence covers only part of this intersection. PowerDynamics supports hidden-voltage estimation, physical recentring, and prospective zero-, one-, and two-source load inference. A retrospective audit links weak-case recovery to profiled manifold distance, while a 120-frame diagnostic studies information accumulation without upgrading it to a recovery experiment. ANDES supports known-source event recognition and documents failure under source hold-out. Multi-family candidate-conditioned inference with state and integrity nuisance still requires one common experimental contract.

III. JOINT STATE, EVENT, AND SOURCE PROBLEM

A. HYBRID DAE AND ELECTRICAL MEASUREMENT MODEL

Let $x(t) \in \mathbb{R}^{n_x}$ denote differential states, $z(t) \in \mathbb{R}^{n_z}$ algebraic states, $u(t)$ scheduled inputs, $q(t)$ network configuration, and $\theta_k = \theta_0 + G_\alpha \alpha_k$ physical parameters with a slowly varying operating coordinate α_k . A physical intervention $a_h(t)$ is indexed by hypothesis

$$h = (c, S, \tau, a, d), \quad h \in \mathcal{H}, \quad (1)$$

where c is event family, S is a bus or branch support, τ is onset, a contains signed severities on that support, and d describes the allowed temporal profile. A single-source event has $S = \{\ell\}$ and scalar severity $\gamma = a_\ell$. The evaluated load contract fixes c and τ and uses

$$S \subseteq \mathcal{L}, \quad K = |S| = \|a\|_0 \leq 2, \quad |\mathcal{L}| = 16, \quad (2)$$

TABLE 1. Capability comparison for the closest methodological families. “Held source” means that the physical asset is absent from source-labeled training or tuning. Entries describe published tasks and assumptions, not a cross-paper performance ranking.

Work	State target	Event/source target	Physics mechanism	Sparse/partial sensing	Held source	Integrity separate
Katanic <i>et al.</i> [3]	Differential and algebraic DAE states	Unknown injectors, not event taxonomy	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [13]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance, not compositional source
Qu <i>et al.</i> [14]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [16]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [17]	Distribution-system state	Line outage	Grid-model-informed graph learning and outlier-resistant SE	Yes	Not stated	No
Moshtagh <i>et al.</i> [18]	Transmission-system state	Topology variation, not source output	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad-data tests
Pandey <i>et al.</i> [4]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [5]	None	Fault detection and location	OLS/CNN with strategic PMU placement	Yes	Not stated	No
Li <i>et al.</i> [29]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [35]	None	Known/unknown event type and location	Siamese distance with learned rejection	Limited/full PMU sets	Open-set rejection	Rejection boundary
Hu and Cheng [36]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No physical-source posterior
Rimkus <i>et al.</i> [6]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	No source labels for imputer	Corruption via imputation
Zhai <i>et al.</i> [7]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmark contracts	Attack target only
This paper	Hidden voltage field; state uncertainty	Eight-label ontology; all load supports with $K \leq 2$	DAE hypotheses; quadratic physical likelihood	Eight prescribed PMU buses	Physical bank avoids source-label head; learned baseline fails holdout	Yes

which gives one no-event, 16 single-source, and $\binom{16}{2} = 120$ two-source hypotheses. The plant is

$$\dot{x}(t) = f(x, z, u, q, a_h; \theta_k) + w_x(t), \quad (3)$$

$$\mathbf{0} = g(x, z, u, q, a_h; \theta_k), \quad (4)$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \theta_k) + b_k] + v_k. \quad (5)$$

Here M_k selects the channels that are available at time k , b_k is a measurement-corruption process, and v_k is measurement noise. A dropout changes M_k ; bad data change b_k ; neither is a physical input to (3)–(4). This separation permits a physical event and a measurement failure to coexist.

The latent variables have deliberately different time scales and meanings. The pair (x, z) is the fast dynamic–algebraic state; α is slow operating-point or parameter drift; a_h is an abrupt, sparse physical intervention; q is a discrete network mode; and (M, b) describes data availability and corruption. A colored model-discrepancy state c_k may enter either the process or measurement residual, as tested in Section V, but is not interchangeable with an event.

Within a smooth mode, assume the DAE is locally index one, so g_z is nonsingular after the reference-angle constraint is imposed. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1} \left(g_x f + g_u \dot{u} + g_\theta \dot{\theta} \right). \quad (6)$$

Equation (6) is not applied across a fault insertion or topology switch, where algebraic variables can jump and the model must be segmented. Differential states remain continuous unless a simulator-specific reset map is declared.

The algebraic network core is

$$I = Y_{\text{bus}}(q, \theta)V, \quad S_i = P_i + jQ_i = V_i I_i^*. \quad (7)$$

Writing $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$ gives

$$P_i = |V_i| \sum_j |V_j| [G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}], \quad (8)$$

$$Q_i = |V_i| \sum_j |V_j| [G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}], \quad (9)$$

where $\vartheta_{ij} = \vartheta_i - \vartheta_j$. A line outage changes the branch primitives and therefore Y_{bus} ; a bus fault adds a shunt-admittance intervention; generation and load changes modify the appropriate active/reactive injection equations. For branch $e = (i, j)$, let t_e be the complex off-nominal tap, y_e the series admittance, and $y_e^{f,sh}, y_e^{t,sh}$ the terminal shunts. Its terminal currents obey

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \quad (10)$$

The PMU operator selects the actual instrumented terminal and orientation. Replacing terminal current with complete

bus-injection current, or changing the tap and charging convention, changes the likelihood.

At the acquisition boundary, native measurements can include V_{abc} , I_{abc} , frequency, ROCOF, quality flags, and availability. Positive-sequence phasors define the balanced network likelihood used here. Negative- and zero-sequence components, when available, remain unbalance or integrity evidence rather than being mixed into the positive-sequence state. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2 \angle V_i}{dt^2}, \quad (11)$$

but an implemented likelihood must use the causal estimator and filtering semantics of the corresponding PMU backend.

B. SPARSE PMUS AND REQUESTED STATE FUNCTION

Both evidence campaigns use PMU buses

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad |\mathcal{P}| = 8. \quad (12)$$

The unobserved set is $\mathcal{U} = \{1, \dots, 39\} \setminus \mathcal{P}$, with $|\mathcal{U}| = 31$. The state-estimation campaign stacks rectangular voltage and controlled terminal-current components,

$$y_{\mathcal{P},k} = \text{col}_{p \in \mathcal{P}} [\Re V_{p,k}, \Im V_{p,k}, \Re I_{p\ell,k}, \Im I_{p\ell,k}], \quad (13)$$

and requests the hidden complex-voltage function

$$z_{\mathcal{U},k} = \text{col}_{i \in \mathcal{U}} [\Re V_{i,k}, \Im V_{i,k}] \in \mathbb{R}^{62}. \quad (14)$$

The event campaign also uses voltage angle, frequency, availability, and, in its longer-window implementation, current and ROCOF. Measurement semantics are kept campaign-specific rather than declaring channels equivalent across simulators.

Figure 2 fixes the network and sensing geometry used by both campaigns. The topology contains 46 physical branches; the colored load buses identify the 16-source candidate set rather than a claim that every event family has been evaluated at those locations. The drawing is explanatory only: all physical responses and electrical-distance features use the audited branch data and admittance model.

C. EIGHT EVENT LABELS AS COMPOSABLE STATES

Table 2 defines the eight abnormal labels in the historical event bank. The scientific variables are not the label numbers. Classes 1–4 are single physical interventions; classes 5 and 7 are measurement-integrity states; classes 6 and 8 are compositions. Class 8 is a prescribed mixed event in the available simulations, not evidence of open-set recognition.

Table 3 states how the four physical families enter the candidate DAE. It defines the operators without implying that the joint posterior has already solved every family.

The internal decision at time k is

$$\left(\hat{z}_{\mathcal{U},k}, \Sigma_{z,k}, \hat{c}_k, \hat{K}_k, \hat{S}_k^{\text{phys}}, \hat{m}_k, \hat{\ell}_k^{\text{meas}}, C_k \right), \quad (15)$$

where m_k denotes the integrity state and $C_k \subseteq \mathcal{H}$ is a credible candidate set. A support is empty when no physical event applies. The candidate set is required because limited observability can make a forced Top-1 decision scientifically unjustified.

D. CORRECTNESS AND EVALUATION UNITS

State accuracy uses the mean hidden-voltage relative complex error (HVRE)

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (16)$$

This trajectory-and-bus aggregate is reported as a percentage. It is TVE-style but is not the standard per-frame synchrophasor TVE conformity statistic. Event detection, event-family macro-F1, exact physical Top-1/Top-3, and integrity-source Top-1 use their own declared populations. The joint correctness indicator is one only when every applicable component of (15) is correct; a correct event family cannot conceal a wrong source. Confidence intervals use an independently generated trajectory, event parent, or rebuilt plant as the statistical unit, never individual 30-Hz samples.

Causal inference permits only $y_{0:k}$ at time k . A fixed-lag smoother, complete-window statistic, or centered temporal feature is retrospective and is labeled accordingly. Source generalization requires that physical source ℓ be absent from all source-labeled training and tuning, not merely that the test trajectory use a new random seed.

IV. RECOVERABILITY AND DIAGNOSABILITY UNDER SPARSE PMUS

A. FUNCTIONAL OBSERVABILITY OF THE REQUESTED VOLTAGE FIELD

Linearizing the index-reduced DAE around a verified equilibrium and sampling at $\Delta t = 1/30$ s gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad (17)$$

$$\delta y_k = C \delta x_k + v_k, \quad (18)$$

$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k. \quad (19)$$

The reduction assumes a regular local algebraic Jacobian after the reference mode is fixed. Here $\delta x_k \in \mathbb{R}^{114}$ contains reduced differential coordinates, C maps them to 32 rectangular voltage/current PMU channels, and $L_{\mathcal{U}}$ maps them to 62 rectangular hidden-voltage components. These matrices are local and hypothesis dependent after a topology change.

For a horizon of N transitions, define

$$\mathcal{O}_N = \text{col} (C, CA_d, \dots, CA_d^N). \quad (20)$$

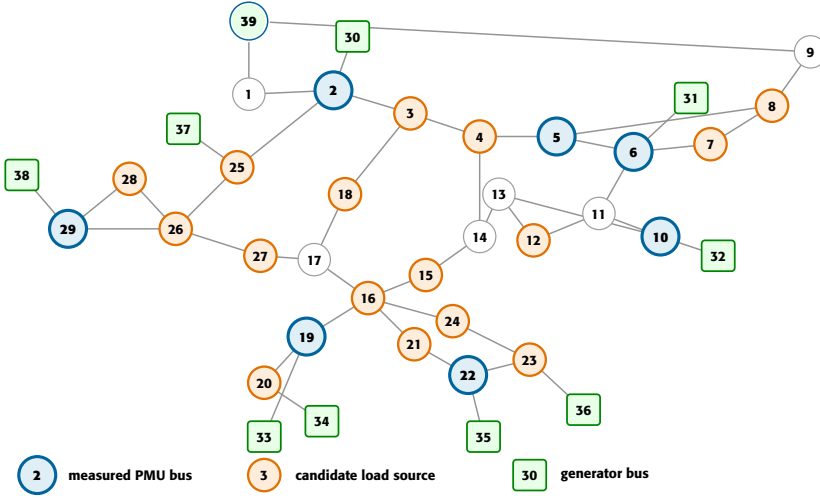
Full-state observability requires $\text{rank}(\mathcal{O}_N) = 114$. The requested voltage at time $r \leq N$ instead has map $L_r = L_{\mathcal{U}} A_d^r$. Recovering the complete local target sequence uses $\mathcal{L}_N = \text{col}(L_0, \dots, L_N)$.

Proposition 1 (Finite-horizon functional reconstruction): For the noise-free local system, the target at time r is uniquely determined by the output sequence through N if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (21)$$

The complete target sequence is uniquely determined if and only if $\ker \mathcal{O}_N \subseteq \ker \mathcal{L}_N$.

Proof: Two initial states produce the same output history exactly when their difference belongs to $\ker \mathcal{O}_N$. Their



Sparse sensing and inference burden

8 of 39 buses measured

$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}$.

16 load-source candidates

Orange nodes define the closed physical bank evaluated in the load study.

31 hidden voltage targets

The network has 46 branches (12 transformers) and 10 generator buses.

Schematic geometry

Responses and electrical distances use the audited Y_{bus} ; drawing length has no electrical meaning.

FIGURE 2. IEEE 39-bus sensing and candidate geometry. Blue circles mark the eight prescribed PMU buses; orange circles mark the 16 load buses in the evaluated physical candidate dictionary; green boxes mark generator buses. Bus 39 carries both a generator and a PMU. The vector drawing preserves the 46-branch connectivity used to form Y_{bus} but is not a geographic map or a proxy for electrical distance.

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. “Physical source” and “integrity source” are separate outputs.

ID	Operational meaning	Physical intervention c	Physical source ℓ	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

TABLE 3. Physical intervention operators in the candidate bank. Algebraic jumps are resolved by the post-event network constraints; differential-state resets are absent unless explicitly stated.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive, $Y_{bus}^+ = Y_{bus}^- + y_f e_\ell e_\ell^T$, for the declared clearing interval	admissible buses	complex fault admittance or resistance/reactance	x continuous; z may jump
Line outage	Set branch status $q_{ij} : 1 \mapsto 0$ and remove its tap, series, charging, and shunt primitives from Y_{bus}	in-service branches	discrete status and switching time	x continuous; z may jump
Generation change	Modify scheduled active/reactive injection or a declared governor reference at generator bus ℓ	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model at load bus ℓ	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

requested outputs at time r agree for every such pair exactly when the difference also belongs to $\ker L_r$. Stacking the maps gives the sequence statement. Equivalently, the rows of the requested map lie in the row space of $\mathcal{O}_N$. ■

This standard functional-observability result [41], [42] fixes the target and time index before any estimator is assessed. It also shows why a low full-state rank alone does not preclude recovery of the hidden-voltage function. In a nonlinear DAE, however, a favorable kernel inclusion at one equilibrium is only first-order evidence: local functional fac-

torization requires a constant-rank neighborhood condition on the feasible observation fibers [43].

The Gauss–Markov specialization adds a variance statement to the rank condition. Stack the output perturbations as $\delta Y_N = \mathcal{O}_N \delta x_0 + v_N$, assume $v_N \sim \mathcal{N}(0, R_N)$ with $R_N \succ 0$, and define $G_N = \mathcal{O}_N^T R_N^{-1} \mathcal{O}_N$.

Corollary 1 (Minimum-variance functional estimate): If (21) holds, the best linear unbiased estimate of the time- r target

and its error covariance are

$$\widehat{\delta z_{\mathcal{U},r}} = L_r G_N^\dagger \mathcal{O}_N^\top R_N^{-1} \delta Y_N, \quad (22)$$

$$\Sigma_{z,r}^* = L_r G_N^\dagger L_r^\top. \quad (23)$$

Every other linear unbiased estimator has covariance $\Sigma_{z,r}^* + Q R_N Q^\top$ for some Q satisfying $Q \mathcal{O}_N = \mathbf{0}$.

Proof: Condition (21) implies $L_r = L_r G_N^\dagger G_N$, which proves unbiasedness of (22). Direct substitution gives (23). Any other unbiased linear map is the displayed map plus Q with $Q \mathcal{O}_N = \mathbf{0}$. The covariance cross terms vanish because $\mathcal{O}_N^\top Q^\top = \mathbf{0}$, leaving the positive-semidefinite term $Q R_N Q^\top$. ■

For whitened measurements, $\text{tr}(\Sigma_{z,r}^*)$ quantifies noise amplification in the requested voltage field. It depends on singular values and target alignment, not rank alone. The numerical curve below is an implementation of this quantity with a tolerance-dependent pseudoinverse; it is a conditioning diagnostic rather than an exact certificate.

B. NUMERICAL AUDIT FOR THE PRESCRIBED EIGHT-PMU CASE

For the anchor-time target used by the structural audit, define

$$\Pi_N = \mathcal{O}_N^\dagger \mathcal{O}_N, \quad (24)$$

$$r_F(N) = \|L_{\mathcal{U}}(I - \Pi_N)\|_F, \quad (25)$$

$$r_{F,i}(N) = \|L_i(I - \Pi_N)\|_F. \quad (26)$$

A zero residual is an exact numerical condition only when the matrix scaling and rank decision are defensible. A nonzero value certifies that the computed observable subspace does not contain every requested anchor-time direction.

The implementation formed $\mathcal{O}_N^\dagger \mathcal{O}_N$ and retained eigenvalues above 10^{-9} of its maximum. Figure 3 turns the audit into a claim test. The pseudo-rank reaches 54 at $N = 30$ but falls to 39 at $N = 180$, although the exact rank of a vertically augmented observability matrix cannot decrease. Gramian squaring, unstable-mode scaling, and a relative threshold therefore make this pseudo-rank unsuitable as an algebraic certificate. Longer history still lowers the trace-style noise bound by more than five orders of magnitude, but $r_F(N)$ remains nonzero. The evidence supports improved numerical conditioning, not full-state or exact target observability. Empirical voltage recovery is evaluated separately and is not used to repair the failed certificate.

Horizons $N \in \{0, 3, 10, 30, 60, 120, 180\}$ correspond to 0–6 s. The nonlinear nominal trajectories contain 91 frames, or 3 s, so $N = 180$ is not a matched causal horizon for those records. Its bus ranking was nevertheless exported before nonlinear scoring and is retained only as a preregistered, cross-horizon numerical diagnostic. The empirical test correlates $r_{F,i}(180)$ with bus-level HVRE; because both derive from the threshold-sensitive projector, that association is exploratory until an SVD-based, noise-tied rank audit is frozen. It does not convert the six-second local audit into a three-second observability proof.

C. EVENT-SOURCE DIAGNOSABILITY FROM CANDIDATE MANIFOLDS

State recoverability and source diagnosability have different nuisance spaces. The construction below specializes established descriptor-model distinguishability [8], matched-subspace nuisance profiling [9], and fault/attack identifiability [45], [48] to candidate-specific DAE responses and explicit integrity supports. The contribution is the common PMU problem contract and its separation from the requested-state problem, not the underlying projection identity. Under candidate h , stack the available channels over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H} \xi_h + \mathcal{E}_{S,H} \beta_h + \varepsilon_H, \quad \varepsilon_H \sim \mathcal{N}(\mathbf{0}, R_H), \quad (27)$$

where $s_{h,H}$ is the intervention response, ξ_h is the locally admissible initial-state deviation, and $\mathcal{E}_{S,H} \beta_h$ is corruption on a fixed support S . Allowing each candidate to refit nuisance variables produces the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{col}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (28)$$

This candidate-specific form is required for topology-changing events because $\mathcal{O}_{i,H}$ and $\mathcal{O}_{j,H}$ need not agree.

Let $\Delta s_{ij} = s_{i,H} - s_{j,H}$ and

$$\mathcal{N}_{ij,S} = [\mathcal{O}_{i,H}, -\mathcal{O}_{j,H}, \mathcal{E}_{S,H}]. \quad (29)$$

The R_H^{-1} -orthogonal nuisance projector and the pairwise separation margin are

$$P_{ij,S}^\perp = I - \mathcal{N}_{ij,S}(\mathcal{N}_{ij,S}^\top R_H^{-1} \mathcal{N}_{ij,S})^\dagger \mathcal{N}_{ij,S}^\top R_H^{-1}, \quad (30)$$

$$D_{ij,S}^{(P)}(H) = \frac{1}{2} \|P_{ij,S}^\perp \Delta s_{ij}\|_{R_H^{-1}}^2. \quad (31)$$

Equation (31) is half the minimum squared weighted distance between the two affine manifolds. It becomes the Gaussian mean-separation term only after nuisance values are fixed at the closest points. If all candidates are conditioned on one common pre-event state, the corresponding model replaces the two independent state blocks in (29) by $\mathcal{O}_{i,H} - \mathcal{O}_{j,H}$. The analysis below uses independent candidate-specific state blocks; it does not use the common-state alternative.

Proposition 2 (Local composite-hypothesis indistinguishability): If $D_{ij,S}^{(P)}(H) = 0$, then $\mathcal{M}_i(S)$ and $\mathcal{M}_j(S)$ intersect. Hence there exist admissible nuisance values under candidates i and j that induce the same Gaussian measurement law. No decision rule can distinguish the pair uniformly over the declared nuisance set; with equal prior mass on those two parameter points, its average error is at least $1/2$.

Proof: The margin is zero exactly when $\Delta s_{ij} \in \text{col}(\mathcal{N}_{ij,S})$, which is equivalent to an intersection of the two affine mean sets. At the intersecting nuisance values, the means and covariance are identical. The observation law is therefore identical under both parameter points, so the two conditional error probabilities sum to one. ■

When the corruption support is unknown but at most q channels are affected, the conservative margin is

$$D_{ij}^{(q)}(H) = \min_{S: |S| \leq q} D_{ij,S}^{(P)}(H). \quad (32)$$

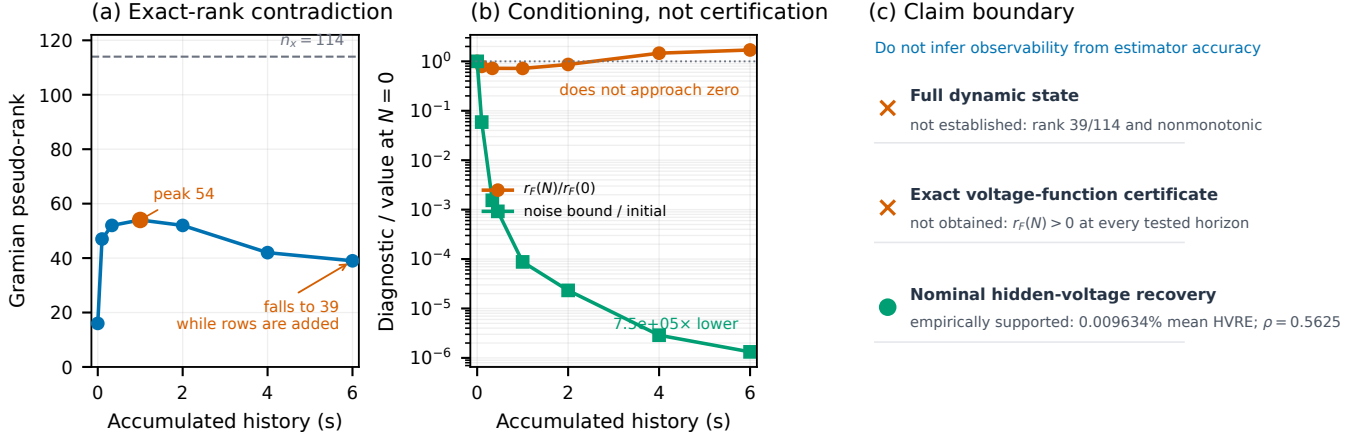


FIGURE 3. What the prescribed eight-PMU audit permits one to claim. (a) The relative-threshold Gramian pseudo-rank decreases after $N = 30$ even though samples are added, contradicting the monotonicity required of exact rank. (b) Values normalized to their $N = 0$ levels separate two effects: accumulated history sharply lowers the trace-style noise bound, whereas the functional residual does not approach zero. (c) Consequently, the audit does not establish the full dynamic state or an exact hidden-voltage function. The small nominal TEST error is empirical evidence under the frozen experiment, not an observability proof.

This union-of-subspaces formulation connects source localization to attack identifiability, sparse strong observability, and unknown-input reconstruction [44], [46]–[48]. It also states the condition required for distinguishability under sparse corruption.

Corollary 2 (Monotonicity under an added PMU): For the same pair, covariance model, nuisance specification, and integrity budget, adding a conditionally independent PMU block cannot reduce the profiled margin:

$$D_{ij,S}^{(\mathcal{P} \cup \{p\})}(H) \geq D_{ij,S}^{(\mathcal{P})}(H). \quad (33)$$

For every nuisance vector, the augmented squared residual is the old nonnegative residual plus the new PMU residual. Taking the minimum preserves the inequality. Equality with a sum of separately profiled margins does not hold in general because the same nuisance variables must explain both sensor blocks; additivity requires fixed simple hypotheses or additional separability conditions.

Observation time has an analogous but distinct role. Align a persistent event at its onset and reduce the regular local DAE to

$$\xi_{k+1} = A_r \xi_k + B_r a, \quad y_k = C_r \xi_k + D_r a, \quad \xi_0 = L_r a. \quad (34)$$

Stacking T samples gives $Y_T = \mathcal{H}_T a$ after the common no-event trajectory is removed. Let $\mathcal{A}_{2K}$ denote vectors with at most $2K$ nonzero entries.

Proposition 3 (Finite-horizon sparse event injectivity): All event vectors with at most K active sources are uniquely determined by the noise-free T -sample output if and only if

$$\ker \mathcal{H}_T \cap \mathcal{A}_{2K} = \{0\}. \quad (35)$$

For a fixed support S , stable amplitude estimation further requires the weighted Gramian $\mathcal{H}_{T,S}^T R_T^{-1} \mathcal{H}_{T,S}$ to be well conditioned, not merely nonsingular.

Proof: Two K -sparse events produce the same history exactly when their difference, which is at most $2K$ sparse, lies

in $\ker \mathcal{H}_T$. This proves the equivalence. On a fixed support, generalized least squares has covariance equal to the inverse support Gramian; its smallest eigenvalue therefore controls noise amplification. ■

For competing supports S and R , define the profiled temporal separation

$$\Delta_{S \rightarrow R}^2(T) = \min_b \|\mathcal{H}_{T,S} a - \mathcal{H}_{T,R} b\|_{R_T^{-1}}^2. \quad (36)$$

Under prefix-consistent causal whitening and a fixed nuisance domain, appending samples cannot reduce (36): each candidate fit acquires a nonnegative residual term before minimization. If the optimally profiled steady signatures differ, the separation grows as $\Theta(T)$; if they coincide and the remaining transient is square summable, it can saturate. Thus a finite-horizon failure may be delay limited, whereas equality of equilibrium signatures is only a necessary warning for persistent ambiguity. Both statements are conditional on the same local model and sensing operator.

The affine margin is first order. If nuisance absorbs an event tangent, relative curvature can still produce a squared distance of order a^4 rather than a^2 under a regular nuisance chart [49]. That possibility is not invoked as the mechanism here: the retrospective load audit classifies every tested direction as first-order resolvable and rejects the simple curvature-dominated delay explanation. A bounded nonlinear remainder is still required before any positive affine margin is promoted to an exact nonlinear certificate.

The executed load-intervention case gives a narrower matched-signal specialization. Let D_g be the whitened unit-amplitude response for candidate source g and let Σ be the normal-data covariance before whitening. The amplitude Fisher information is

$$\mathcal{I}_g = D_g^T \Sigma^{-1} D_g, \quad |a|_{\min} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{\mathcal{I}_g}}, \quad (37)$$

where the second expression is the local amplitude required by a one-sided Gaussian score with false-alarm probability α and power $1 - \beta$. The formula predicts detectability, not source identity.

For source j as the closest single-signature alternative, the information left after its amplitude is fitted is

$$\mathcal{I}_{g \rightarrow j} = \mathcal{I}_g - \frac{(D_g^\top \Sigma^{-1} D_j)^2}{D_j^\top \Sigma^{-1} D_j} = \mathcal{I}_g(1 - \mu_{gj}^2), \quad (38)$$

with μ_{gj} the covariance-weighted signature coherence. Equation (38) is the profiled pairwise Fisher information for one signature against another. It need not predict multicandidate confusion once amplitude priors, unequal alternatives, and finite nonlinear response are present.

For a calibrated source posterior, define the smallest candidate set satisfying

$$\mathcal{C}_k(\alpha) = \arg \min_{\mathcal{C} \subseteq \mathcal{H}} \left\{ |\mathcal{C}| : \sum_{h \in \mathcal{C}} p(h | y_{0:k}) \geq 1 - \alpha \right\}. \quad (39)$$

A diagnosability-aware system abstains from exact Top-1 when $|\mathcal{C}_k| > 1$ or when its closest competing margin falls below a development-set threshold. The load-intervention study evaluates exact support and a retrospective profiled-distance mechanism but does not fit initial-state nuisance $\mathcal{O}_h \xi_h$ or corruption nuisance $\mathcal{E}_S \beta_h$. Candidate-set calibration, prospective resolution delay, and PMU loss or addition still require a source-disjoint experiment.

V. PHYSICS-BASED STATE ESTIMATION AND EVENT-SOURCE INFERENCE

A. EXACT TARGET AND A GENERATIVE CANDIDATE MODEL

Let $\zeta_k = L(x_k, z_k)$ be the electrical quantity requested by the application. Under squared loss, the exact Bayesian target is

$$\hat{\zeta}_k^{\text{MMSE}} = \mathbb{E}[\zeta_k | y_{0:k}], \quad \hat{h}_k^{\text{MAP}} = \arg \max_{h \in \mathcal{H}} p(h | y_{0:k}). \quad (40)$$

This statement is an optimality definition, not an implemented-algorithm claim.

For candidate $h = (c, \ell, \tau, \gamma, d)$, assume $x_0 \sim \mathcal{N}(\bar{x}_0, P_0)$, Gaussian process and measurement noise, the hard constraint $g_h(x_t, z_t) = \mathbf{0}$, and a Laplace prior on active corruption coefficients. With

$$r_{h,t} = M_t[y_t - h_{\mathcal{P}_t}(x_t, z_t) - b_t], \quad (41)$$

the negative twice-log joint density conditional on h , up to constants and the declared corruption-support prior, is

$$\begin{aligned} \mathcal{J}_h^\circ = & \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \|r_{h,t}\|_{R_t^{-1}}^2 \\ & + \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + 2\lambda_b \sum_{t=0}^k \|b_t\|_1, \end{aligned} \quad (42)$$

subject to the DAE constraints. Crucially, $\mathcal{J}_h^\circ$ does not contain the hypothesis prior. The physical family and source

enter through F_h, g_h , and the network/measurement operators rather than as an asset-label head.

The evidence is the constrained integral

$$p(y_{0:k} | h) = \int_{g_h=\mathbf{0}} \exp[-\mathcal{J}_h^\circ/2] dX dZ dB. \quad (43)$$

A Laplace approximation on locally identifiable directions gives

$$p(h | y_{0:k}) \propto p(h) \det^+(H_h)^{-1/2} \exp[-\mathcal{J}_h^\circ(\hat{X}_h, \hat{Z}_h, \hat{B}_h)/2], \quad (44)$$

so the prior appears exactly once. The pseudo-determinant requires the active constraint and corruption support to be fixed locally. If a Huber loss, a soft DAE penalty, or an omitted determinant is used without a corresponding normalized density, the result is a penalized evidence score, not a posterior probability.

The learned component is deliberately restricted. The earlier engineered feature system can propose a candidate set or learn a correction $\psi_\phi(r_{h,0:k})$ to physical log evidence. It must consume candidate residuals, not source identity alone, and must be tested on held-out physical sources. The required ablation is learned-only, physics-only, physics plus learned discrepancy, and the complete diagnosability-aware rule.

B. EVALUATED BAYESIAN LOAD-SOURCE MODEL

One restricted instance of the candidate model is executed rather than proposed. For each of 16 load buses, a centered finite difference of the nonlinear time-domain simulator forms a 30-frame, 32-channel response dictionary,

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \quad \epsilon = 0.005. \quad (45)$$

The dictionary is physical: it is regenerated by changing the load at candidate bus g and simulating the network. No source classifier or source label is fitted. Symmetric simulations also estimate self-curvature Q_g and pairwise interaction Q_{ij} . For support S and amplitudes a_S , the frozen mean is

$$\mu_S(a) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j \in S \\ i < j}} a_i a_j Q_{ij}. \quad (46)$$

The cross term is not a cosmetic refinement: it represents the mixed derivative of the nonlinear event manifold and is isolated experimentally.

After subtracting the normal mean, $r = \mu_S(a) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$. The W2 covariance combines the frozen channel covariance with temporal AR(1) coefficient 0.3512. It was estimated from equilibrium records with synthetic measurement noise, not ambient load trajectories. Instrument and timing error, noise sweeps, and PMU loss are absent.

The cardinality prior is $p(K = 0, 1, 2) = (0.20, 0.50, 0.30)$; supports are uniform conditional on K , and active amplitudes are independent $\mathcal{N}(0, 0.05^2)$. The normalized support evidence is

$$Z_S = \int \mathcal{N}(r; \mu_S(a), \Sigma) \prod_{i \in S} \mathcal{N}(a_i; 0, 0.05^2) da_S, \quad (47)$$

with $Z_\emptyset = \mathcal{N}(r; \mathbf{0}, \Sigma)$ and

$$p(S | r) \propto Z_S \frac{p(|S|)}{\binom{16}{|S|}}. \quad (48)$$

Operational inference uses 31-node Gauss–Hermite quadrature per active amplitude. An independent two-dimensional Gauss–Kronrod reference checks representative supports. The posterior yields $p(K | r)$, exact-support ranks, per-bus inclusion probabilities, and conditional amplitude intervals. A model-averaged amplitude also requires an explicit zero atom when the bus is inactive; its interval calibration remains partial and is not used as a primary claim.

The prospective bank evaluates all 137 supports at the known 30-frame onset. A separate 120-frame dictionary preserves exact prefixes at $T \in \{30, 60, 90, 120\}$ and is used only for physical-manifold, information, and runtime diagnostics. This executed model conditions on the nominal initial state, assumes no corrupted channels, and uses fixed topology. It tests the sparse-support specialization of Proposition 3; it does not implement the state/integrity nuisance fit in (27) or validate the multi-family posterior in (44).

C. NOMINAL ESTIMATOR HIERARCHY

The estimator sequence changes one information source at a time. B0 retains only the equilibrium. B1 adds instantaneous PMU conditioning, B2 adds causal state propagation, and B3 tests retrospective temporal information. When the equilibrium becomes stale, E06-H estimates a physically admissible operating point through regularized nonlinear recentering. The full nonlinear Hybrid-DAE fixed-lag posterior is not evaluated in the experiments.

All nominal estimators use the same equilibrium y_0 , hidden-voltage center $z_{\mathcal{U},0}$, sampled dynamics A_d , PMU map C , and hidden-output map $L_{\mathcal{U}}$. This shared model isolates the value of measurement updating and temporal propagation.

The equilibrium baseline B0 returns

$$\hat{z}_{\mathcal{U},k}^{\text{B0}} = z_{\mathcal{U},0}. \quad (49)$$

It measures how much of the score can be obtained by remaining at the nominal power-flow solution.

The snapshot estimator B1 computes a prior-conditioned linear minimum-mean-square-error (LMMSE) update independently at every frame,

$$K_{\text{LMMSE}} = P_0 C^\top (C P_0 C^\top + R)^{-1}, \quad (50)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0), \quad \hat{z}_{\mathcal{U},k}^{\text{B1}} = z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{\delta x}_k^{\text{B1}}. \quad (51)$$

B1 uses network-constrained cross-covariance but no state carried from frame $k - 1$.

Conditional on one linear-Gaussian candidate with no corruption, B2 is not an ad hoc module: the Kalman recursion computes the exact conditional mean and covariance in (40). Candidate uncertainty, nonlinear switching, unknown integrity support, and non-Gaussian discrepancy are what make the enclosing hybrid problem approximate.

The causal Kalman estimator B2 applies the classical linear-Gaussian recursion [55]:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1}, \quad (52)$$

$$P_{k|k-1} = A_d P_{k-1|k-1} A_d^\top + Q, \quad (53)$$

$$\nu_k = (y_k - y_0) - C \hat{x}_{k|k-1}, \quad (54)$$

$$S_k = C P_{k|k-1} C^\top + R, \quad (55)$$

$$K_k = P_{k|k-1} C^\top S_k^{-1}, \quad (56)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k. \quad (57)$$

The covariance update uses the Joseph form. The hidden estimate is $z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{x}_{k|k}$. B2 is causal because only $y_{0:k}$ enters the result at k .

B3 is an exact fixed-lag Gaussian smoother, following the linear Gaussian backward-recursion principle in [56], evaluated at lags $\ell \in \{0, 1, 3, 5, 10, 15, 30, 60\}$ frames. It uses measurements through $k + \ell$ to refine the estimate at k and is therefore retrospective by $\ell/30$ seconds. Dense and cached implementations agree numerically on representative lags. B3 tests whether temporal information beyond the causal update improves the hidden-voltage target; it is not used to support a real-time claim.

D. MARGINAL CALIBRATION AND COLORED DISCREPANCY

For the hidden two-component voltage error at bus i , B2 provides

$$\Sigma_{i,k} = L_i P_{k|k} L_i^\top. \quad (58)$$

The original covariance is $\mathbf{U}_0$. A two-block temperature model $\mathbf{U}_2$ is fitted on an independent calibration split:

$$\Sigma_{i,k}^{\mathbf{U}_2} = T \Sigma_{i,k} T^\top, \quad T = \text{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}). \quad (59)$$

The temperatures minimize Gaussian negative log likelihood (NLL) on calibration data and are frozen before TEST. Coverage, NLL, normalized innovation squared (NIS), and temporal innovation correlation are reported separately; nominal marginal coverage alone is not called complete calibration.

The residual study considers a latent Gauss–Markov process

$$c_{k+1} = F_c c_k + \eta_k. \quad (60)$$

In D2-M, c_k enters the PMU measurement residual through a calibration-fitted low-rank basis. In D2-P, it enters a sparse physical process dictionary. Basis, rank, F_c , and latent covariance are selected using calibration records only. The experiment asks whether the observed low-rank structure improves reconstruction, hidden uncertainty, or innovation whiteness. It does not assume that a lower innovation autocorrelation is sufficient.

E. MODEL-ADEQUACY STATISTIC AND RECENTRING ABLATIONS

Let μ_ν and Σ_ν be the mean and shrunk covariance of B2 innovations on the nominal calibration set. The per-frame adequacy statistic is the five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^\top \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (61)$$

Mismatch labels and hidden truth do not enter (61); hidden truth is used only after inference to label frames whose hidden HVRE exceeds the frozen evaluation threshold. AUROC and AUPRC therefore measure whether observed innovations anticipate unacceptable hidden error, not whether the detector identifies a mismatch family.

Three variants diagnose the failure mechanism without merging their evidence contracts. R1 replaces the nominal equilibrium with the true plant equilibrium and is explicitly oracle. R2-A estimates an eight-dimensional grouped observed-bus voltage offset using a 60-frame trailing window, nominal matrices, ridge regularization, and only PMU measurements. It is a causal negative ablation: its center/state cross-covariance is neglected and it does not enforce the nonlinear AC equations.

The corrected regularized physical estimator addresses M6 operating-point mismatch at one equilibrium snapshot. Let

$$d = \text{col}(\Delta P_{PQ}, \Delta Q_{PQ}, \Delta P_{PV}) \in \mathbb{R}^{43} \quad (62)$$

collect active and reactive changes at 17 admissible PQ loads and active-power changes at nine PV buses. Voltage is parameterized as $V(\theta, \rho) = \exp(\rho + j\theta)$. Defining the scaled PMU residual as $e_y(V) = D_y^{-1}[h_P(V) - y_P]$, the estimator solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \mathcal{J}(V, d), \quad (63)$$

$$\mathcal{J}(V, d) = \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \frac{\lambda}{q_d} \|d\|_2^2.$$

where h_P is the audited voltage/terminal-current operator, g_{AC} includes PQ balance, explicit PV-magnitude constraints, solved PV reactive power, and bus 31 as Slack, and D_y scales voltage and current components separately. The weights are $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$, selected on a disjoint DEV split; continuation initialization is then frozen for TEST. Neither the true nuisance vector nor the hidden voltages enter the TEST solve.

Because d may be non-unique, evaluation distinguishes nuisance information from target information. If J_{yd} and J_{zd} are the observed-PMU and hidden-voltage Jacobians with respect to d after eliminating voltage through the AC constraints, the local target residual is

$$r_{d \rightarrow z} = \frac{\|J_{zd} N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)}, \quad (64)$$

where $N(J_{yd})$ spans the PMU-uninformed nuisance nullspace. Exact target identifiability would require a zero residual. The experiment instead tests the weaker empirical question: whether (63) reconstructs the hidden voltage accurately despite rank-deficient nuisance information, and whether its Laplace covariance is credible.

VI. VALIDATED BENCHMARK AND EXPERIMENTAL PROTOCOL

A. BENCHMARK PROVENANCE AND NUMERICAL GATES

The primary plant uses Julia 1.11.9, PowerDynamics.jl 5.0.0, and NetworkDynamics.jl 1.3.0 [57]. It contains 39 buses, 46

branches, 19 loads, 10 sixth-order Sauer–Pai machines, nine AVR Type-I controllers, and nine TGOV1 governors. The descriptor realization has 192 coordinates: 114 differential and 78 algebraic. These parameters are package-defined tutorial data. We therefore classify the plant as a numerically validated pedagogical benchmark rather than an authoritative representation of an external grid.

The validation ladder checks the model before estimator error is scored. A same-plant static translation was audited after correcting tap orientation, shunts, line charging, and full- Y_{bus} injection accounting. The largest remaining static discrepancy is 1.71×10^{-3} MVA in terminal flow. The controlled PMU operator agrees with the simulator to 2.71×10^{-7} pu in complex voltage and 1.43×10^{-14} pu in analytic/backend terminal current.

Dynamic validation checks the equilibrium, local modes, nonlinear trajectories, and linearization. The DAE equilibrium residual is 4.66×10^{-13} . After removing the reference mode, the least-damped physical pair is $-0.09818 \pm j0.42948 \text{ s}^{-1}$. Nonlinear Rodas5P simulations remain finite for the audited perturbations. Amplitude sweeps satisfy

$$\|y_{\text{NL}}(\epsilon) - y_{\text{LIN}}(\epsilon)\|_2 \propto \epsilon^p, \quad (65)$$

with fitted slopes between 1.982 and 2.000 for representative state, governor-reference, and load perturbations, consistent with first-order linearization error.

B. NOMINAL RECONSTRUCTION AND CALIBRATION

The nominal dataset contains 20 development and 100 test trajectories, each sampled at 30 Hz for 3 s. Perturbations are normal-operation changes to governor references, loads, or their combination. The equilibrium, linear model, $P_0 = 10^{-2}I$, $Q = 10^{-6}I$, and $R = 10^{-6}I$ are frozen before TEST. B0, B1, B2, and all B3 lags are evaluated on the same trajectories. A separate 30-trajectory calibration split fits U1/U2 temperatures and low-rank discrepancy variants; TEST is not used for selection.

Hidden truth is used only for post-hoc metrics. The eight observed buses are excluded from every primary reconstruction score. The phasor prediction is formed as $V_0 + \Delta V$, and angle error is wrapped using $\text{atan2}(\sin \Delta\vartheta, \cos \Delta\vartheta)$. These rules correct an earlier exploratory implementation that incorrectly interpreted $\angle \Delta V$ as the absolute angle; no superseded result from that implementation is retained.

C. PHYSICAL MISMATCH CAMPAIGN

For each mismatch case, a fresh PowerDynamics plant is rebuilt from copied source tables, followed by power flow, dynamic initialization, and nonlinear simulation. The original package CSVs remain hash-identical. The seven families are network $R/X/B$ (M1), machine parameters (M2), governor parameters (M3), AVR parameters (M4), ZIP/load model (M5), operating-point redispatch (M6), and coupled mismatch (M7). Stress scale is $m \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$. These are engineering stress-test ranges, not probability priors.

TABLE 4. Frozen experimental contracts. State and event campaigns retain separate simulators, splits, targets, and denominators.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families, seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 candidate load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source–amplitude–noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; 9,600 weak + 800 no-event and 8,000 finite-amplitude TEST records	Source–amplitude–noise record
Prospective 137-hypothesis load inference	Fresh H_0 + 16 singles + 120 pairs; 3,488 physical trajectories; 10,564 noise realizations	Physical trajectory for simulation uncertainty; complete record for decisions
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 trajectories and 2,400 noise realizations	Retrospective source-pair trajectory
Long-horizon likelihood diagnostic	Exact prefixes through 120 frames; 32 nominal pair trajectories; all 137 hypotheses	Diagnostic trajectory; not a population recovery test
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied; exact source	303 decisions from 101 event parents $\times$ 3 seeds

All 980 main-grid cases and all 175 refinement cases passed power-flow, initialization, and time-domain acceptance. B1, B2, U0, U2, and the adequacy score are frozen from nominal data. Bootstrap intervals use seeds as independent units. The plots read the frozen per-case table directly; a previously used refined breakpoint mixed aggregation definitions and is excluded.

D. ORACLE, OFFSET, AND PHYSICAL-MAP ADAPTATION CONTRACTS

The oracle subset compares the frozen estimator with true-equilibrium recentering and true local relinearization on the same mutated trajectories. Oracle quantities are evaluation-only. The offset-only recentering study uses new DEV seeds 101–110 and TEST seeds 201–220, disjoint from all earlier datasets. Its window, cadence, regularization, and activation threshold are selected on DEV and frozen before TEST. No event labels, hidden state, or mismatch family enter the R2-A estimate.

The corrected regularized physical-recentering campaign is a separate M6-only contract. It uses 30 fresh DEV cases from seeds 601–610 and 60 TEST cases from seeds 611–630, with 20 TEST equilibria at each of $m \in \{0.5, 1.0, 1.5\}$. DEV selects the nuisance scale, regularization, and continuation initialization from a small fixed grid. TEST uses only the observed voltage and terminal-current phasors; hidden voltages are revealed afterward for scoring. A solution is accepted only when the nonlinear optimizer succeeds, all outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator; the frozen rejection table is empty. A six-case subset is also solved with an independent constrained optimizer, and its partial convergence status is reported rather than upgraded to full solver equivalence.

All numerical tables and empirical figures are regenerated from the hashed snapshot under `generated/frozen`. The build script checks the sample counts, acceptance flags, rank-deficient nuisance information, and preregistered adaptation-

closure range before producing the tables and figures.

E. LOAD-SOURCE INFERENCE CONTRACT

The load-source campaign uses the PowerDynamics plant and the same eight PMU buses, but it has its own frozen normal banks, event dictionary, and test manifests. The dictionary covers 16 candidate load buses listed in its manifest. Centered $\pm 0.5\%$ nonlinear simulations define (45); the corresponding dictionary hash is fixed before the Bayesian tests. Candidate source and amplitude are inferred from the PMU window. The true source and true amplitude appear only in the evaluator.

V1 tests finite changes of ± 1 , ± 2.5 , ± 5 , and $\pm 7.5\%$ on 2,560 event and 320 no-event records. V2 expands normal calibration to 1,000 records, uses 500 normal and 640 event records for likelihood selection, then opens two TEST blocks. The weak block contains 9,600 events at magnitudes from 0.01 to 0.4% and 800 no-event records. The finite block contains 8,000 events from ± 1.25 to $\pm 9\%$. Noise seeds are disjoint across CAL, DEV, and TEST. A test record is one source–amplitude–noise realization; repeated samples within its 30-frame window are never counted as independent cases.

W2 and L2 are selected on CAL/DEV, respectively, and frozen for both V2 TEST blocks. The weak-event operating threshold is 0.5 posterior event probability. No TEST row enters covariance estimation, likelihood selection, prior selection, or threshold tuning.

The subsequent GLOBAL-137 confirmation freezes W2, D , Q , Q_{ij} , priors, onset, and quadrature before generating new data. Its support space contains H_0 , all 16 singles, and all 120 pairs. Four weak–weak magnitude combinations, one weak–strong, one moderate, and one finite combination are evaluated under all sign patterns; four additional magnitudes cover single events. All 3488 native nonlinear trajectories solve successfully. Three disjoint noise realizations per event path plus 100 H_0 records produce 10564 decisions. The GLOBAL-137 identifiers have zero overlap with 1,816 historical manifest identifiers. Replayed noise is not promoted to an independent

physical-system replicate.

The exact weak-regime study is explicitly retrospective. It replays the frozen GH31 likelihood over 2,400 physical paths at horizons 5–30 and asks which geometry predicts cardinality and exact-support recovery; it does not refit the estimator. The T120 contract regenerates a true-time-local numerical dictionary through 120 post-event frames and validates 32 pair trajectories at the nominal operating point. Its information, equilibrium, and runtime outputs are diagnostics. Independent operating-point validation and an all-term analytic variational continuation remain pending, so T120 is not a frozen recovery claim.

The load campaigns cover one physical family, fixed topology, and known onset. They do not validate missing PMUs, corrupted channels, cross-simulator transfer, or the state-conditioned multi-family posterior.

F. EVENT-DIAGNOSIS CAMPAIGN AND SIGNAL EVIDENCE

The separate ANDES campaign [58] uses the same eight PMU buses and 138 source configurations: one normal, 39 faults, 34 outages, ten generation, 18 load, eight missing-data, ten generation-and-missing, eight bad-data, and ten mixed configurations. Five replicas give 690 trajectories. Deterministic source/replica formulas vary severity over $\{0.06, 0.09, 0.12, 0.15\}$, operating scale over $\{0.97, 0.985, 1.00, 1.015, 1.03\}$, and the simulator/noise seed; source and 2-s onset stay fixed. Per-PMU AR(1) Gaussian noise is calibrated from normal, available RAW0001 rows. Trajectories last about 5 s; missing/corrupted intervals last 0.7 s. Physical interventions act in the simulator, while integrity changes act on sampled measurements.

Three replicas per source configuration enter training, one development, and one test, producing 414/138/138 complete-trajectory splits. Consequently, the primary event benchmark evaluates new replicas at sources already represented in training. The separate asset-disjoint audit removes the physical source from all source-labeled fitting and tuning. Results from these contracts are reported side by side but never interpreted as the same generalization task.

The historical baseline is a hierarchy of extremely randomized trees. Its 136-feature version uses trailing voltage magnitude, angle, frequency, missingness, and a normalized disturbance score. The 448-feature version adds multi-scale statistics, adaptive residuals, and simultaneous contrasts among PMUs. A deterministic availability gate handles directly observed dropout. These are physics-guided features and typed decision heads; they are not a DAE-constrained learning architecture, and the ensemble's leaf-vote scores are not treated as calibrated probabilities.

Figure 4 uses the first labeled episode of six representative families in the 89.65-min RAW0001 reference simulation. The curves are measured simulator outputs, not sketches or smoothed reconstructions. They show why no single channel suffices: a fault creates a short voltage sag, an outage redistributes current, injection changes have slower fre-

quency/current responses, and measurement failures require availability or phase-consistency information.

VII. NOMINAL RECONSTRUCTION AND STRUCTURAL EVIDENCE

A. WHAT THE SPARSE PMUS RECOVER

Table 5 gives the nominal TEST results. Conditioning the equilibrium estimate on the current PMU sample (B1) reduces hidden-voltage HVRE by more than one order of magnitude. The causal Kalman recursion (B2) improves on B1 in 69.0% of the same trajectories, but its paired mean advantage is only -0.000270 percentage points. Most of the nominal gain therefore comes from the measurement-conditioned physical map; temporal propagation supplies a smaller correction.

Figure 5(a) makes the trajectory distributions and scale separation visible, while panel (c) shows the paired differences rather than comparing overlapping marginal intervals. The equilibrium baseline is more than one order of magnitude worse than either measurement-updated estimator. This is not a comparison against learned or nonlinear DAE estimators; it is a controlled decomposition of equilibrium centering, instantaneous PMU conditioning, and causal temporal propagation within one local model.

B. FUNCTIONAL STRUCTURE AS A PREDICTOR OF ERROR

At horizon 180, the finite-horizon Gramian has pseudo-rank 39 of 114 and the aggregate target residual is nonzero; neither full-state nor exact functional observability is declared. The preregistered bus residual nevertheless correlates with B2 HVRE at Spearman $\rho = 0.5625$ [Fig. 5(b)], with a two-sided permutation p -value of 0.0012. Because the 31 buses share trajectories and network coupling, bus-resampling intervals are treated as exploratory rather than as independent-unit confidence intervals. The same qualitative relation appears for B1.

The relationship is informative but incomplete. Bus 36 has the largest structural residual, whereas the five largest B2 errors occur elsewhere. The projector captures one source of difficulty but not the nonlinear excitation distribution, estimator covariance, or every conditioning effect. Electrical distance to the nearest PMU has not been computed on the same audited network, so the analysis cannot yet show that $r_{F,i}$ adds information beyond that confounder. The residual remains a diagnostic, not a binary observability certificate.

C. RETROSPECTIVE SMOOTHING TEST

Across lags from one frame to 2 s, the implemented fixed-lag smoother never improves mean HVRE over lag zero; every positive lag is slightly worse in the nonlinear TEST trajectories. A correctly specified linear-Gaussian smoother cannot increase expected squared error, so this result diagnoses a violated modeling premise rather than a general failure of smoothing. The local model absorbs deterministic governor/load excitation into white process noise, while lag-one innovation autocorrelation remains 0.602 and the Ljung–Box test rejects whiteness. An unknown-input estimator [12] and

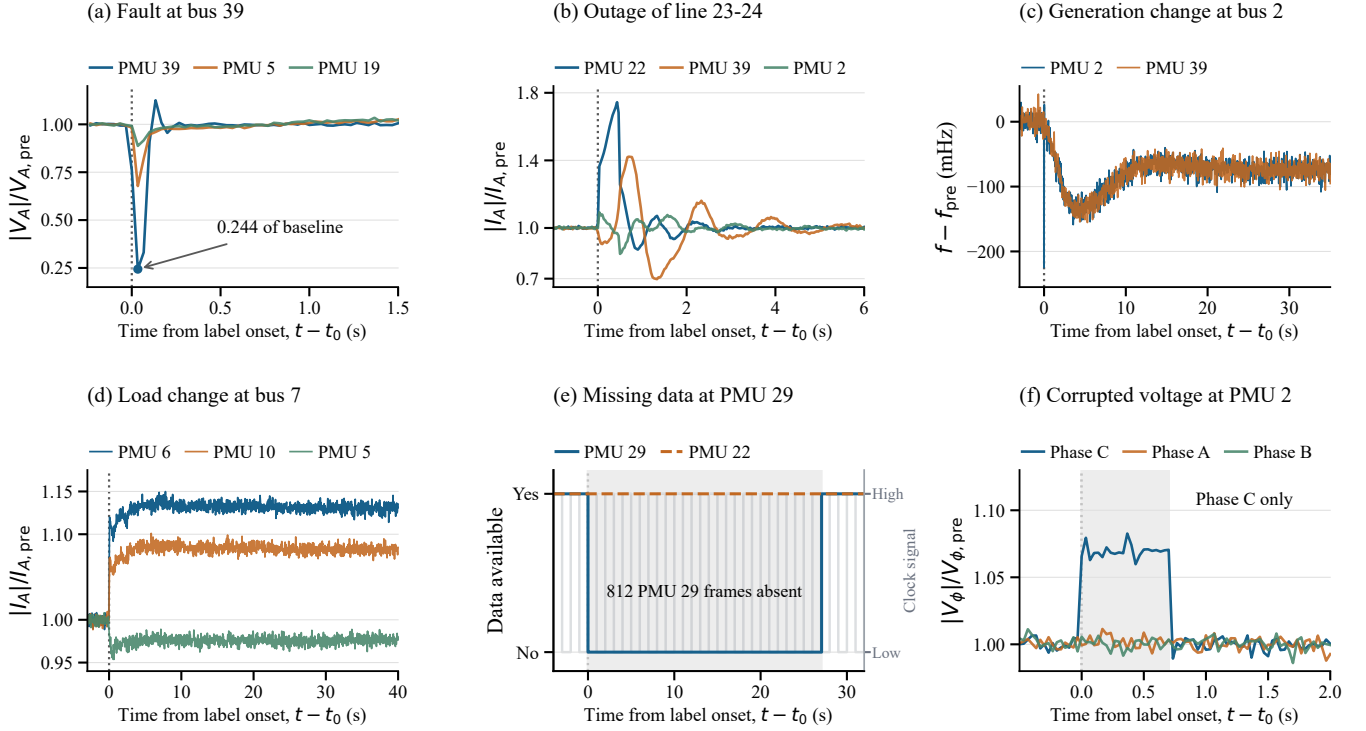


FIGURE 4. Representative measured event signatures from RAW0001, sampled at approximately 30 frames/s. (a) Bus fault, (b) line outage, (c) generation change, (d) load change, (e) missing data, and (f) single-phase corruption. The 812-frame missing interval in (e) is approximately 27.1 s; the full-height light-gray square wave and right-hand axis show a schematic reference clock derived from the recorded time coordinate, not a received PMU channel. Thus time advances while PMU 29 packets are absent. Dotted lines mark the first labeled sample; gray shading marks missing or corrupted intervals. Magnitudes are normalized by the preceding 3-s median, frequency is relative to its corresponding median, and time scales differ by panel. The plotted measurements and source-file hashes are frozen in the evidence snapshot.

TABLE 5. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^{\circ}$)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

a re-estimated process model are required before a latency–accuracy conclusion is drawn.

VIII. UNCERTAINTY, PHYSICAL MISMATCH, AND ADAPTATION

A. MARGINAL CALIBRATION DOES NOT WHITEN INNOVATIONS

The nominal B2 covariance is conservative. Two temperatures fitted on 30 independent CAL trajectories bring TEST marginal coverage closer to 95% and improve hidden NLL, but do not change lag-one innovation autocorrelation or the rejection of whiteness. Four principal components explain 99.2% of calibration-residual variance, yet a rank-four measurement Gauss–Markov correction worsens HVRE and undercovers both voltage components; process placement is neutral. Marginal calibration, temporal adequacy, and useful discrepancy structure are therefore distinct properties. The complete diagnostics remain generated from the frozen arti-

facts but are omitted here to preserve space for the prospective event results.

B. MISMATCH RESPONSE DEPENDS ON MODEL FAMILY

The rebuilt-plant campaign reveals a clear family dependence (Fig. 6). At the largest tested scale, network mismatch raises the median error by $16.99\times$ and an operating-point shift raises it by $8.32\times$. Machine, governor, AVR, and load-model perturbations remain near the nominal median because this 3-s excitation does not measurably expose those parameter changes. Their near- $1\times$ ratios are regime-limited null responses, not evidence of robustness. The tested estimator fails visibly when the algebraic network map or equilibrium center changes.

Nominal covariance repair does not transfer automatically. Averaged across all mismatch cases, U0 has 92.9% 95% coverage and nominally fitted U2 has only 47.5%. The reduction reflects physical shift, not a calibration/test leak: U2

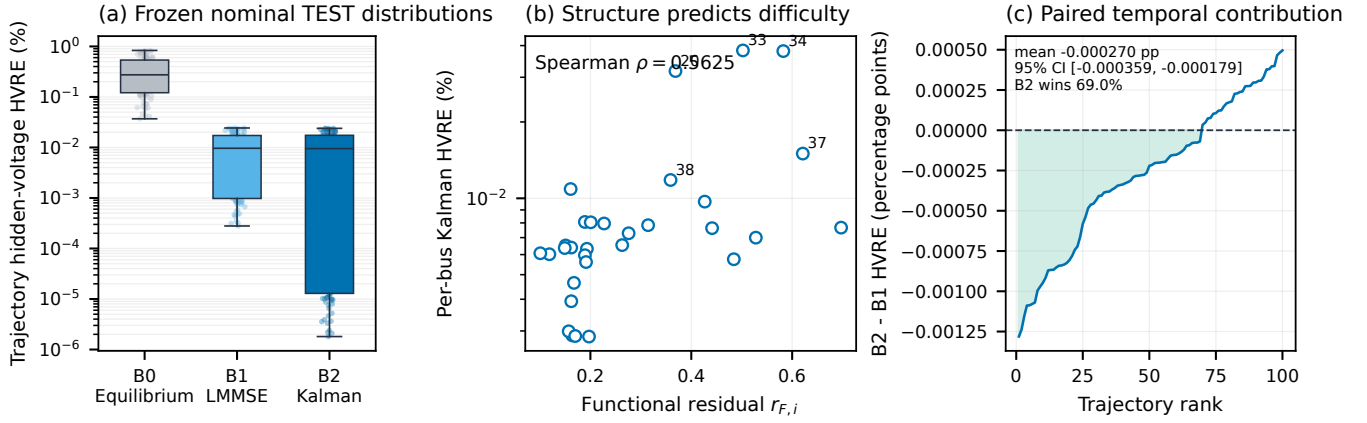


FIGURE 5. Nominal reconstruction and structural evidence. (a) Trajectory-level hidden-voltage HVRE distributions on the same 100 nonlinear TEST trajectories. (b) Per-bus causal Kalman HVRE versus the horizon-180 functional residual exported before TEST scoring; labels identify the five buses with largest empirical error. (c) Sorted paired B2-minus-B1 HVRE differences; negative values favor the causal estimator, and the annotation reports the paired bootstrap interval and win rate.

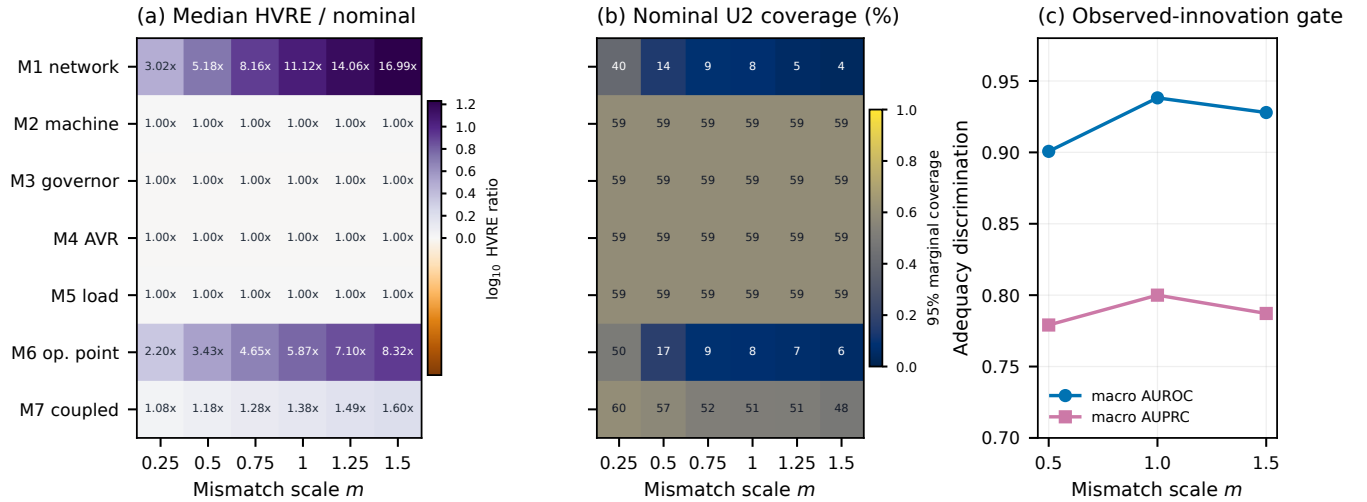


FIGURE 6. Physical-mismatch atlas for the fixed nominal estimator. (a) Median B2 hidden-voltage HVRE relative to the pooled $m = 0$ median for every family-scale cell; annotations show the untransformed ratio and color encodes its base-10 logarithm. (b) Mean nominal-U2 marginal 95% coverage under the same rebuilt plants. (c) Macro adequacy discrimination from observed innovations; undefined family-level AUROCs are omitted from the macro mean. Each family-scale cell contains 20 independently rebuilt plants.

remains frozen from nominal CAL data. This result limits the interpretation of Section V: nominal dispersion scaling is not robustness to a changed plant.

C. ADEQUACY DETECTION AND PHYSICAL RECENTERING

The observed-PMU innovation statistic detects many cases in which hidden HVRE is unacceptable. At the largest tested scale, its macro AUROC/AUPRC is 0.928/0.787 [Fig. 6(c)]. Family-level AUROC is omitted when every frame has the same frozen adequacy label; AUPRC remains defined. The statistic can flag a stale model, but it neither identifies the mismatch family nor supplies a calibrated alarm probability.

The earlier R2-A offset-only ablation remains scientifically useful because it isolates what does not solve the problem. Across nonnominal identifiable M1, M6, and M7 cases it closes only 0.4% of the oracle gap, despite passing its nominal-safety gate. That failure does not imply that adapta-

tion from observed PMUs is impossible; it shows that grouped voltage offsets without AC balance and a rebuilt physical center are inadequate.

The regularized-recentering experiment tests that distinction for M6. Every TEST case satisfies the prespecified convergence gate. Across the three mismatch scales, the physical solve removes 97.4%–98.3% of the error gap to an oracle center [Fig. 7(a)–(b)]. The median solve takes 61.7 ms. Table 6 retains the point errors, paired intervals, and tail runtime; these static timings do not establish end-to-end real-time operation.

The nuisance information matrix has median rank 25 among 43 coordinates. Even with that nullspace, the requested voltage field is recovered with two orders of magnitude less error than the frozen center. This supports target recovery despite nuisance non-uniqueness. It does not prove exact functional identifiability because the local hidden-voltage

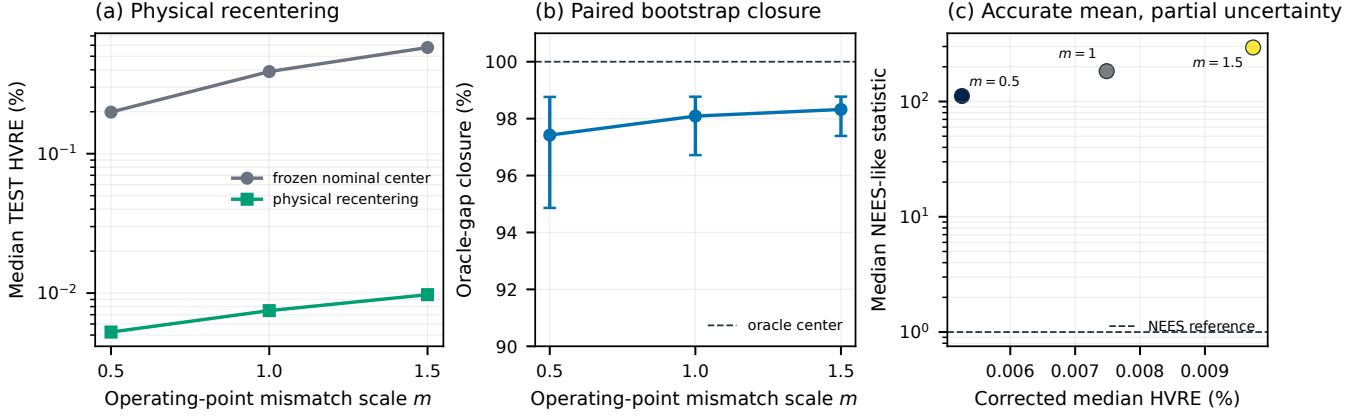


FIGURE 7. Corrected regularized physical recentering for held-out M6 operating-point mismatch. (a) Median hidden-voltage HVRE for the nominal center and the physical solve. (b) Paired oracle-gap closure with 95% bootstrap intervals over 20 independent TEST cases per scale; the oracle-center reference is 100%. (c) Corrected point error versus the median joint NEES-like diagnostic. Marginal 95% coverage is 100% at each scale, but the joint statistic remains far above its nominal reference.

TABLE 6. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

residual remains nonzero, nor does it show that the injection vector is recovered.

Point accuracy does not certify posterior uncertainty. The pseudoinverse Laplace covariance is formed only on locally informed nuisance directions; because $J_{zd}N(J_{yd}) \neq 0$ in (64), it does not represent the image of the PMU-uninformed nuisance nullspace in the hidden-voltage target. This explains how marginal intervals can cover every component while the joint NEES-like statistic grows with mismatch and remains far above its nominal reference [Fig. 7(c)]. The independent constrained solver also converges on only part of its small diagnostic subset. M6 point reconstruction is supported; joint uncertainty, network mismatch, coupled mismatch, and sequential adaptation are not.

D. PROSPECTIVE LOAD CARDINALITY AND SUPPORT RECOVERY

The single-source development path establishes the components used by the larger support test. The centered numerical tangent differs from either one-sided derivative by at most 0.235%, W2 yields NIS/frame 0.998 with lag-one correlation -0.0026, and the fitted nonlinear remainder has order 2.00 with bootstrap interval [1.79, 2.24]. The 137-hypothesis confirmation then freezes D , Q , Q_{ij} , W2, priors, onset, and GH31 before generating new trajectories.

All 3488 physical trajectories succeed, and none of their identifiers overlaps the 1,816 historical identifiers checked. Table 7 reports 10564 complete-record decisions. Weak-weak events are the limiting regime: cardinality accuracy is 37.4%, exact-support Top-1 is 24.4%, and Top-3 is 59.8%.

Weak-strong Top-1 rises to 66.2%, while moderate and finite Top-1 reach 98.8% and 100.0%. These are support decisions over all 120 pairs, not two independent single-source classifications.

The weak transition isolates what improves first. As the larger weak magnitude rises from 0.033% to 0.121%, mean $p(K = 2)$ increases from 0.088 to 0.758, whereas exact-support Top-1 remains 24.4% [Fig. 8(b)]. The model can therefore accumulate evidence for two active sources while remaining unsure which pair produced the response. Over all event records, detection AUROC/AUPRC is 0.990/0.9999, with 7.3% false negatives at the 0.5 threshold. The observed H_0 false-positive count is zero, but only 100 H_0 records were exposed; its 95% upper bound is 3.6%, so this is not a continuous-alarm guarantee. Per-bus inclusion has AUROC 0.950 and ECE 0.029, but recall 0.645 shows that calibrated-looking marginal scores do not repair weak exact-support recovery.

The retrospective audit supplies a mechanism without changing the frozen estimator. Across 2,400 weak-regime cases, global profiled distance correlates with exact support at $\rho = 0.780$, compared with $\rho = 0.140$ for the support-only quantity [Fig. 8(c)]. All tested local directions are first-order resolvable; the curvature-dominated explanation and simple a^{-2} or a^{-4} resolution-delay laws are not supported. This negative result replaces the earlier mismatched comparison between pairwise Fisher information and pooled 16-way confusion.

The T120 diagnostic narrows the structural interpretation. The within-artifact global separation proxy grows monoton-

TABLE 7. Fresh prospective 137-hypothesis confirmation. The bank contains H_0 , 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top- k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K = 2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak-weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak-strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

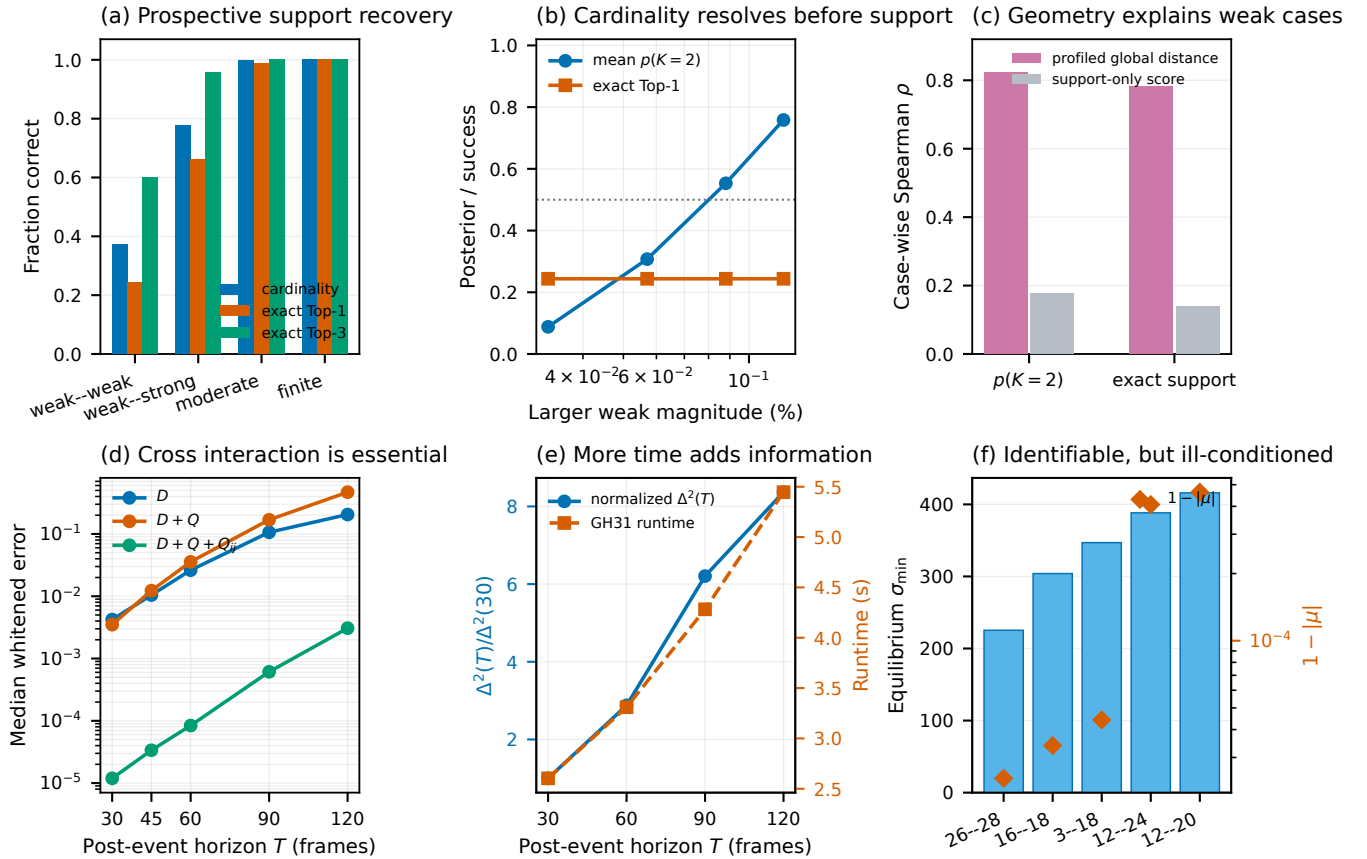


FIGURE 8. From prospective recovery to its information boundary. (a) Frozen GLOBAL-137 cardinality and exact-support results. (b) Within weak-weak cases, cardinality evidence grows while exact Top-1 remains flat. (c) Retrospective-only mechanism audit: the global profiled manifold distance predicts $p(K = 2)$ and exact recovery more strongly than a support-only score. (d) Nominal-operating-point T120 validation: pairwise interaction Q_{ij} is necessary to track simultaneous nonlinear responses. (e) The within-artifact separation ratio grows through 120 frames, alongside measured GH31 runtime; absolute separation values are withheld because their scale is not reconciled with the earlier profiled metric. (f) The five smallest equilibrium singular values remain positive but accompany near-unit signature coherence. Panels (a)–(b) are prospective primary evidence; (c) is retrospective; (d)–(f) are contract diagnostics, not recovery claims.

ically and is 8.37 times its T30 value at T120; $\gamma_{4,\infty} = 2844.44 > 0$, and no equilibrium pair is numerically degenerate in the frozen local model. This does not establish universal source identifiability. The hardest pair, buses 26–28, has positive $\sigma_{\min} = 225.3$ but coherence 0.999976; buses 7–12 have coherence -0.997947. Distinct signatures can therefore be severely ill-conditioned.

Physical validation also reveals why the mixed term matters. At T120, the median whitened response error is 0.2053 for D , 0.4690 for $D + Q$, and 0.0031 for $D + Q + Q_{ij}$. For the complete quadratic model, median error relative to the

nearest margin is 5.36e-04 and the maximum is 0.088. The 32-path validation bank varies pair, sign, and amplitude but not operating point, so its status remains limited. Full 137-hypothesis GH31 takes 5.446 s at T120 on the benchmark; this is computational evidence, not a real-time claim. Independent operating-point validation and scale reconciliation are required before the T120 contract is frozen for prospective recovery.

E. KNOWN-SOURCE DETECTION AND LOCALIZATION

On the known-source TEST split, the availability-gated hierarchy reaches detection F1 0.989, event macro-F1 0.905, physical Top-1/Top-3 0.592/0.719, and integrity Top-1 0.974. Expanding the representation adds 9.6 percentage points to physical-source ranking over the compact hierarchy. Multiscale dynamics, adaptive residuals, and PMU contrasts help when the source is represented during training; the availability rule contributes mainly to integrity handling.

The gap between detection F1 and physical Top-1 confirms that a recognizable disturbance need not have a uniquely recoverable source. False alarms decrease to 1.45 episodes per normal-labeled minute, but the normal exposure is only about 7.15 min per split. That rate is evidence of alarm burden in a short synthetic stream, not an operational false-alarm guarantee.

F. SOURCE HOLDOUT BREAKS LEARNED LOCALIZATION

With the physical source excluded from source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. Detection and event-family confusion have been removed from this test. The remaining error shows that the learned location heads and candidate descriptors retain source-specific associations, invalidating a zero-shot-source claim for the legacy system.

Transfer to RAW0001 is also weak: the basic candidate model localizes only 0.3 of five physical episodes on average. The simulator bank omits some channels used by the longer-window implementation, so no single feature family can be blamed for the deficit. Internal simulation accuracy does not establish cross-record reliability.

Supervised, physics-guided features work for detection and known-source diagnosis but not for source-generalizable inference. The physical load experiment supplies a contrasting result: all single and double supports are evaluated through simulated responses rather than learned asset labels. Its success is restricted to one event family, fixed topology, known onset, and a matched simulator; it does not repair the broader held-source or cross-record failure.

IX. DISCUSSION AND LIMITATIONS

The contracts expose different parts of the inverse problem. Eight voltage/current PMUs recover the requested 31-bus voltage field accurately in the PowerDynamics state tests even though the 114-state local model is not fully observable. On the same simulator, a physical load-response dictionary supports normalized posterior evidence over all zero-, one-, and two-source supports. The ANDES bank measures what the earlier supervised representation retained when source labels were removed. Low hidden-voltage HVRE still does not guarantee source separation, and correct cardinality does not guarantee exact support.

The local theory explains this split. Proposition 1 removes unobservable directions that do not affect the requested voltage function, and Corollary 1 attaches a covariance to that estimable function. Proposition 3 instead asks whether any $2K$ -

sparse event difference lies in the finite-horizon kernel and distinguishes uniqueness from conditioning. Proposition 2 then adds state and integrity nuisance. Large faults, topology switches, and unmodeled controls may violate these local descriptions.

Three theory-to-data links now have different evidential status. The functional residual correlates with bus-level voltage error, but its projector fails rank monotonicity and needs an SVD/noise-tied recomputation. In the prospective event test, weak cardinality becomes confident before exact support. Retrospectively, profiled nonlinear distance is strongly associated with exact weak-case recovery, whereas support-only geometry is weak. The tested directions are first-order resolvable, so the curvature-dominated mechanism is not the explanation for this bank. The T120 equilibrium dictionary has positive subset margin, yet near-unit coherence exposes poor conditioning; the correct conclusion is finite-information ambiguity under the validated local model, not absence of structural ambiguity in general.

The operating-point study also separates target recovery from parameter recovery. Regularized physical recentering closes 98.1% of the oracle-center gap on M6 although only 25 of 43 nuisance directions are locally informed. The hidden-voltage target can therefore be accurate while the injection coordinates remain non-unique. This result is confined to an equilibrium snapshot. It does not cover line outages, faults, or sequential event-conditioned estimation, and the local curvature covariance remains under-dispersed in the joint NEES-like statistic.

The legacy event model supplies one reusable mechanism and one warning. Multiscale summaries, adaptive residuals, and simultaneous PMU contrasts improve known-source physical ranking by 9.6 percentage points. Availability belongs in the integrity state because a missing measurement is observed directly. Source identities and endpoint descriptors, however, allow memorization. Exact localization falls to 1.0% under complete physical-source holdout. The load experiment avoids an asset-label head by integrating evidence for every physical support. Learned residual features may still help, but their contribution must be measured against this physics-only likelihood on the same event parents.

The principal evidence gap is a multi-family end-to-end experiment. The studies have different targets or simulator semantics and are simulated. The Bayesian result covers one family, fixed topology, known onset, closed candidates, matched operation, and synthetic noise; it does not fit state or integrity nuisance. The T120 physical check contains only 32 nominal-operating-point trajectories, and its absolute separation scale is not reconciled with the earlier profiled metric. The ANDES records have fixed onsets and about 7.15 min of normal exposure per split. Neither field validity nor operational false-alarm control is established.

A decisive follow-up first completes independent-operating-point T120 validation and reconciles the separation metric, then freezes a prospective horizon study at 30–120 frames. The multi-family stage should compare learned-

only, physics-only, learned-residual, and diagnosability-aware methods under one audited plant and PMU operator. Event parents and sources must remain disjoint across fitting, calibration, and test. Shift, topology, PMU availability, and long normal exposure must report joint correctness, candidate-set coverage, resolution delay, false alarms per hour, HVRE, calibration, runtime, and failed solves. Until then, multi-family superiority remains unproved.

X. CONCLUSION

Sparse-PMU monitoring contains inverse problems with different information limits. Functional observability determines whether the requested hidden-voltage function is recoverable. Sparse event injectivity and candidate-manifold separation determine whether event cardinality and support remain distinguishable over a finite horizon, including after state and integrity nuisance are fitted. A positive structural margin is not the same as a well-conditioned decision.

The experiments show the distinction. The causal Power-Dynamics estimator attains 0.009634% mean hidden-voltage HVRE, and physical recentering removes 97.4%–98.3% of the operating-point error gap. In the fresh 137-hypothesis load test, exact-support Top-1 rises from 24.4% for weak-weak pairs to 100.0% for finite pairs; weak cardinality can be resolved before source identity. Retrospective profiled distance explains weak-case recovery, while the T120 diagnostic finds positive but nearly collinear equilibrium signatures. The separate ANDES model still yields only 3 correct decisions among 303 after complete physical-source holdout.

Physical candidate responses now have direct evidence for simultaneous interventions in one matched event family. Independent-operating-point horizon validation must precede extension to faults, outages, generation changes, integrity events, changing PMU sets, and simulator shift. The central consequence is practical: state error, event cardinality, exact support, and source diagnosability must be measured separately before a joint decision is trusted.

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